

Reinforcement Learning for Autonomous Aircraft Landing in Crosswinds

Autonomous Flight Control Using Proximal Policy Optimization (PPO)

Project Area: Artificial Intelligence for Aerospace Applications

PROJECT REPORT

Author: Amaraneni Vinitha

Degree: B.Tech Aeronautical Engineering

1. ABSTRACT

Aircraft landing is one of the most critical phases of flight and becomes significantly more challenging in the presence of crosswinds and atmospheric disturbances. Crosswind conditions can cause runway centerline deviations, unstable approach trajectories, excessive sink rates, and hard landings, thereby increasing operational risk and pilot workload. Traditional flight control systems typically rely on fixed control laws and tuning parameters that may not adapt effectively to rapidly changing environmental conditions.

This project investigates the application of Reinforcement Learning (RL) for autonomous aircraft landing under varying crosswind and turbulence conditions. A custom aircraft landing simulation environment was developed using Python and Gymnasium to model vertical descent dynamics, lateral runway tracking, crosswind disturbances, turbulence effects, and touchdown behavior. The environment provides continuous observations of aircraft states while allowing continuous flight control through elevator, throttle, aileron, and rudder commands.

A Proximal Policy Optimization (PPO) agent was trained using Stable-Baselines3 to learn optimal landing strategies through interaction with the environment. The reward structure was designed to encourage stable descent profiles, accurate runway centerline tracking, smooth touchdown behavior, and safe landing performance while penalizing excessive lateral deviation, unstable approaches, and crash events.

Performance evaluation was conducted through Monte Carlo simulations under both nominal and previously unseen wind conditions. Experimental results demonstrated a 100% landing success rate under training wind conditions with a 99% smooth landing rate. Under unseen wind conditions, the trained policy maintained a 90% landing success rate while preserving acceptable touchdown characteristics and lateral tracking accuracy.

Comparative analysis against a heuristic baseline controller showed substantial improvements in touchdown smoothness and overall landing quality.

The results demonstrate the feasibility of Reinforcement Learning as a promising approach for future autonomous flight control systems and intelligent landing assistance technologies in aerospace applications.

2. INTRODUCTION

2.1 Background

Aircraft landing is widely recognized as one of the most demanding and safety-critical phases of flight. During landing, pilots and flight control systems must simultaneously manage aircraft attitude, descent rate, runway alignment, airspeed, and environmental disturbances while operating within narrow safety margins. Even minor deviations during the final approach and flare phases can result in hard landings, runway excursions, structural damage, or loss of aircraft control.

Among various environmental disturbances, crosswinds represent one of the most significant operational challenges. A crosswind is a wind component acting perpendicular to the runway direction. Strong crosswinds generate lateral drift that can push the aircraft away from the runway centerline, requiring continuous corrective actions from either the pilot or the flight control system. In addition, atmospheric turbulence introduces unpredictable disturbances that further complicate the landing process.

Commercial aircraft pilots undergo extensive training to handle crosswind landings safely. However, even experienced pilots face difficulties under highly dynamic environmental conditions. Consequently, the development of autonomous systems capable of assisting or performing landing operations under adverse weather conditions remains an important research area within aerospace engineering.

2.2 Motivation

Traditional aircraft autopilot systems rely primarily on classical control techniques such as Proportional-Integral-Derivative (PID) controllers and gain-scheduled control laws. While these methods have proven effective in many operational scenarios, their performance often depends heavily on accurate system modeling and extensive tuning procedures.

As environmental conditions become increasingly uncertain, traditional control approaches may struggle to maintain optimal performance across a wide range of operating conditions. In particular, handling stochastic disturbances such as varying crosswinds and turbulence requires adaptive decision-making capabilities that are difficult to achieve using fixed control strategies.

Recent advances in Artificial Intelligence and Reinforcement Learning have demonstrated remarkable success in solving complex sequential decision-making problems.

Reinforcement Learning enables an autonomous agent to learn optimal control policies through interaction with its environment without requiring explicit programming of control rules. This capability makes Reinforcement Learning particularly attractive for aerospace applications involving uncertainty, nonlinear dynamics, and dynamic operating conditions.

The motivation behind this project is to investigate whether Reinforcement Learning can learn robust landing strategies capable of maintaining safe touchdown performance under varying crosswind conditions while outperforming traditional heuristic control approaches.

2.3 Project Overview

This project presents the development of a Reinforcement Learning-based autonomous aircraft landing framework. A custom aircraft landing environment was designed to simulate critical landing dynamics, including vertical descent, lateral runway tracking, crosswind disturbances, and turbulence effects.

The learning agent observes aircraft states including altitude, vertical speed, pitch angle, throttle setting, lateral position, lateral velocity, wind estimate, and runway distance. Based on these observations, the agent generates continuous flight control commands consisting of elevator, throttle, aileron, and rudder inputs.

The PPO algorithm was selected due to its stability, sample efficiency, and widespread adoption in continuous control problems. Through repeated interactions with the environment, the agent learns policies that maximize landing safety, touchdown quality, and runway tracking accuracy.

The trained policy is evaluated using extensive simulation experiments and compared against a conventional heuristic controller to assess performance improvements.

3. PROBLEM STATEMENT

Safe aircraft landing under crosswind conditions remains a challenging control problem due to the combined effects of environmental disturbances, nonlinear aircraft behavior, and strict operational safety requirements. Crosswinds introduce lateral drift forces that can move the aircraft away from the runway centerline during the final approach and touchdown phases. Simultaneously, turbulence can generate unpredictable disturbances that affect aircraft stability and descent performance.

Conventional flight control systems typically rely on fixed control laws that are designed for expected operating conditions. While these systems perform effectively under nominal circumstances, they may exhibit degraded performance when exposed to highly variable wind conditions or previously unseen disturbances.

The primary challenge addressed in this project is the development of an autonomous landing controller capable of learning adaptive landing strategies directly through interaction with a simulated environment. The controller must continuously adjust flight control inputs to compensate for crosswinds, maintain stable descent trajectories, minimize touchdown errors, and avoid unsafe landing conditions.

The objective is to determine whether Reinforcement Learning can provide a robust and adaptive alternative to traditional control approaches while maintaining high landing success rates and superior touchdown quality.

4. PROJECT OBJECTIVES

4.1 Primary Objective

The primary objective of this project is to develop an autonomous aircraft landing controller using Reinforcement Learning that can safely land an aircraft under varying crosswind and turbulence conditions.

4.2 Specific Objectives

The specific objectives of the project are as follows:

1. Develop a custom aircraft landing simulation environment capable of modeling approach, flare, touchdown, and runway tracking behavior.
2. Incorporate realistic environmental disturbances including crosswinds and turbulence effects.
3. Design an observation space that captures critical aircraft state information relevant to landing control.
4. Implement a continuous action-space controller capable of generating elevator, throttle, aileron, and rudder commands.
5. Train a Reinforcement Learning agent using the Proximal Policy Optimization algorithm.
6. Design a reward function that encourages safe and efficient landing behavior.

7. Evaluate policy performance through Monte Carlo simulations over a large number of randomized landing scenarios.
8. Compare Reinforcement Learning performance against a conventional heuristic baseline controller.
9. Assess policy robustness under previously unseen wind conditions.
10. Demonstrate the applicability of Reinforcement Learning for autonomous aerospace control systems.

5. LITERATURE REVIEW

5.1 Traditional Aircraft Landing Control

Aircraft landing automation has historically relied on classical control techniques such as PID controllers, Linear Quadratic Regulators (LQR), and gain-scheduled autopilot systems. These methods utilize mathematical models of aircraft dynamics to generate control commands that stabilize the aircraft and guide it toward the runway.

PID controllers remain widely used due to their simplicity and ease of implementation. However, their performance often depends heavily on controller tuning and may degrade under changing operating conditions. Advanced methods such as Model Predictive Control (MPC) have been proposed to improve landing performance by optimizing future control actions while satisfying system constraints. Although MPC can achieve high levels of performance, it typically requires accurate dynamic models and significant computational resources.

5.2 Reinforcement Learning

Reinforcement Learning is a branch of machine learning in which an agent learns optimal behavior through interaction with an environment. Unlike supervised learning, Reinforcement Learning does not require labeled datasets. Instead, the agent receives rewards based on the quality of its actions and gradually learns policies that maximize long-term cumulative reward.

Recent advances in Deep Reinforcement Learning have enabled the successful application of RL to complex continuous control problems including robotic manipulation, autonomous driving, game playing, and aerospace guidance systems.

5.3 Proximal Policy Optimization

Proximal Policy Optimization (PPO), introduced by Schulman et al., is one of the most widely used policy-gradient reinforcement learning algorithms. PPO improves training stability by limiting the magnitude of policy updates through a clipped objective function.

Advantages of PPO include:

- Stable training behavior
- Efficient policy learning
- Compatibility with continuous control problems
- Reduced sensitivity to hyperparameter selection
- Strong empirical performance across diverse environments

Due to these characteristics, PPO has become a popular choice for aerospace control applications involving continuous action spaces.

5.4 Reinforcement Learning in Aerospace Applications

Recent research has explored the application of Reinforcement Learning in various aerospace domains including autonomous guidance, spacecraft attitude control, collision avoidance, trajectory optimization, and unmanned aerial vehicle navigation.

Several studies have demonstrated that Reinforcement Learning can learn control policies capable of adapting to uncertain environments while achieving performance comparable to or exceeding traditional control methods. These findings suggest that Reinforcement Learning represents a promising direction for future autonomous flight systems and intelligent aerospace control architectures.

6. SYSTEM ARCHITECTURE

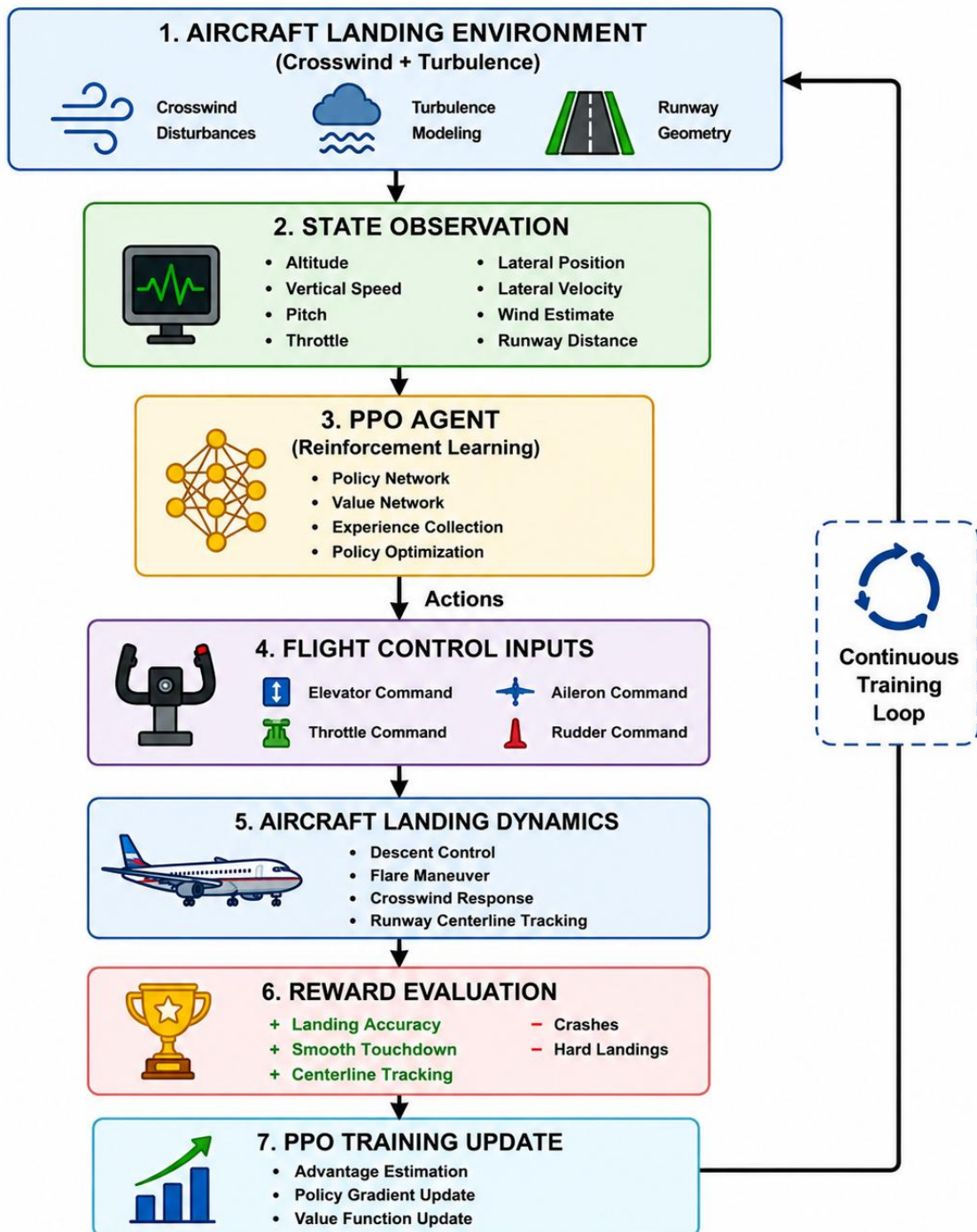
6.1 Overview

The autonomous landing framework developed in this project follows a closed-loop Reinforcement Learning architecture in which a learning agent continuously interacts with a simulated aircraft landing environment. The architecture consists of six major components: the aircraft landing environment, state observation module, PPO agent, flight control inputs, aircraft landing dynamics model, and reward evaluation mechanism.

At each simulation timestep, the environment provides a set of observations describing the current aircraft state. Based on these observations, the PPO agent generates continuous control commands that influence the aircraft dynamics. The resulting aircraft behavior is evaluated through a reward function that quantifies landing performance. This reward signal is then used to update the policy during training.

The overall architecture enables the agent to learn landing strategies through trial-and-error interactions while continuously improving its performance over successive training episodes.

Figure 1. Reinforcement Learning Architecture for Autonomous Aircraft Landing



6.2 Aircraft Landing Environment

The aircraft landing environment represents the operational domain in which the learning agent interacts. The environment models key aspects of the landing process including

altitude dynamics, descent rate behavior, lateral runway tracking, crosswind disturbances, turbulence effects, and touchdown conditions.

Environmental disturbances are randomized across episodes to expose the agent to a wide range of operating conditions and improve policy robustness.

The environment serves as the primary source of observations and reward signals during training.

6.3 State Observation Module

The observation module provides the agent with information describing the current aircraft state.

The observation vector consists of:

- Altitude
- Vertical Speed
- Pitch Angle
- Throttle Position
- Lateral Position
- Lateral Velocity
- Wind Estimate
- Runway Distance

These variables capture the most important information required for autonomous landing decision-making.

6.4 PPO Agent

The Reinforcement Learning controller is implemented using the Proximal Policy Optimization algorithm.

The PPO agent consists of two neural network components:

Policy Network

The policy network determines which control action should be executed based on the current aircraft state.

Value Network

The value network estimates the expected future reward associated with the current state and assists in policy optimization.

Together, these networks enable stable and efficient learning.

6.5 Flight Control Inputs

The agent controls the aircraft using four continuous flight control commands:

Elevator

Controls aircraft pitch attitude and descent behavior.

Throttle

Controls engine thrust and energy management.

Aileron

Controls roll motion and assists with runway alignment.

Rudder

Provides yaw control and crosswind compensation.

The continuous action-space formulation allows smooth and realistic control behavior.

6.6 Aircraft Landing Dynamics

The aircraft landing dynamics module updates aircraft states based on the selected control actions and environmental disturbances.

The dynamics model simulates:

- Vertical descent
- Flare behavior
- Crosswind response
- Runway tracking
- Touchdown performance

The resulting aircraft state is returned to the observation module, completing the control loop.

6.7 Reward Evaluation

The reward function evaluates the quality of the landing maneuver.

Positive rewards are provided for:

- Stable descent
- Accurate runway tracking

- Smooth touchdown

Negative rewards are assigned for:

- Excessive lateral deviation
- Unsafe sink rates
- Hard landings
- Crashes

The reward signal guides policy learning throughout training.

7. ENVIRONMENT DESIGN

7.1 Environment Overview

A custom aircraft landing environment was developed using the Gymnasium framework.

The environment was designed specifically for Reinforcement Learning research and provides a simplified but representative model of aircraft landing behavior under crosswind conditions.

The environment follows the standard Gymnasium interface and supports reset and step operations required by modern Reinforcement Learning algorithms.

Each episode begins with the aircraft positioned at a predefined altitude and continues until touchdown or episode termination.

7.2 Observation Space

The observation space represents the information available to the learning agent.

The observation vector contains eight continuous variables.

Table 1. Observation Space Variables

Variable	Description
Altitude	Aircraft height above runway
Vertical Speed	Current descent rate
Pitch Angle	Aircraft pitch attitude
Throttle Position	Engine thrust setting
Lateral Position	Distance from runway centerline
Lateral Velocity	Sideward motion
Wind Estimate	Estimated crosswind intensity
Runway Distance	Remaining landing distance

These variables collectively describe the aircraft's flight condition and landing state.

7.3 Action Space

The action space consists of four continuous control commands.

Table 2. Action Space Definition

Control Input	Purpose
Elevator	Controls pitch
Throttle	Controls thrust
Aileron	Controls roll
Rudder	Controls yaw

The continuous control formulation allows the agent to learn smooth and realistic flight control behavior.

7.4 Episode Initialization

At the beginning of each episode:

- Altitude is initialized to 400 meters.
- Vertical speed is initialized to a stable descent condition.
- Aircraft pitch is initialized to zero.
- Throttle is initialized to a moderate cruise setting.
- Lateral position is randomized.
- Wind disturbances are randomized.

Randomized initialization improves policy generalization.

7.5 Episode Termination Conditions

An episode terminates when:

Successful Landing

The aircraft reaches runway altitude with acceptable touchdown conditions.

Hard Landing

Touchdown occurs but with degraded landing quality.

Crash

Unsafe touchdown conditions occur.

Time Limit

Maximum simulation duration is reached.

8. AIRCRAFT LANDING DYNAMICS

8.1 Overview

The aircraft dynamics model simulates the evolution of aircraft states during the landing process.

The model captures two major motion components:

1. Vertical Dynamics
2. Lateral Dynamics

These simplified dynamics provide sufficient complexity for Reinforcement Learning experimentation while maintaining computational efficiency.

8.2 Vertical Dynamics

Vertical dynamics determine aircraft altitude and descent rate.

Three primary forces influence vertical motion:

Gravity

Acts downward and continuously increases descent tendency.

Lift

Generated through pitch adjustments.

Thrust

Produced by engine power settings.

The vertical speed update can be expressed as:

$$V_s(k + 1) = V_s(k) + Gravity + Lift + Thrust$$

The altitude update is given by:

$$Altitude(k + 1) = Altitude(k) + V_s(k + 1)$$

These equations model the aircraft descent process throughout the landing maneuver.

8.3 Lateral Dynamics

Lateral dynamics determine runway tracking performance.

Crosswind disturbances generate lateral drift that moves the aircraft away from the runway centerline.

The lateral speed update is influenced by:

- Roll Commands

- Rudder Commands
- Wind Disturbances
- Turbulence
- Damping Effects

The lateral dynamics equation is approximated by:

$$LatSpeed(k + 1) = LatSpeed(k) + Roll + Rudder + Wind + Turbulence$$

The lateral position update is given by:

$$LatPos(k + 1) = LatPos(k) + LatSpeed(k + 1)$$

These dynamics enable realistic crosswind compensation behavior.

8.4 Touchdown Classification

Landing outcomes are classified into three categories.

Smooth Landing

A touchdown satisfying:

- Low sink rate
- Small lateral error

Hard Landing

Touchdown occurs safely but exceeds ideal performance limits.

Crash

Touchdown exceeds acceptable safety limits.

This classification framework enables objective performance evaluation.

9. DISTURBANCE MODELING

9.1 Importance of Disturbance Modeling

Environmental disturbances play a critical role during aircraft landing operations.

To ensure robust policy learning, the environment incorporates stochastic crosswind and turbulence effects.

These disturbances expose the agent to diverse operating conditions during training.

9.2 Crosswind Model

Crosswinds are modeled using a random Gaussian process.

The crosswind disturbance is defined as:

$$Wind \sim N(0, \sigma_w)$$

where:

- N denotes a normal distribution.
- σ_w represents wind intensity.

Wind conditions vary between episodes to improve policy robustness.

9.3 Turbulence Model

Turbulence is modeled using an independent Gaussian disturbance process.

$$Turbulence \sim N(0, \sigma_t)$$

where:

- σ_t controls turbulence intensity.

Turbulence introduces short-term disturbances that challenge aircraft stability and control performance.

9.4 Domain Randomization

To prevent policy overfitting, environmental disturbances are randomized across episodes.

Randomization improves generalization and enables successful operation under previously unseen wind conditions.

10. REWARD FUNCTION DESIGN

10.1 Overview

The reward function represents the most critical component of Reinforcement Learning training.

The objective of the reward function is to encourage behaviors that lead to safe, accurate, and smooth landings while discouraging unsafe flight conditions.

10.2 Runway Tracking Reward

The agent receives rewards for maintaining proximity to the runway centerline.

The penalty term is proportional to lateral deviation:

$$Reward_{lat} = -k_1 | LatPos |$$

This encourages accurate runway alignment throughout the landing maneuver.

10.3 Stable Descent Reward

A target descent profile is enforced through vertical speed penalties.

$$Reward_{vs} = -k_2 |V_s - V_{target}|$$

This encourages stable approach trajectories and reduces excessive sink rates.

10.4 Control Smoothness Reward

Large control inputs are penalized to encourage smooth and realistic aircraft behavior.

Penalties are applied to:

- Elevator Commands
- Roll Commands
- Rudder Commands

This prevents aggressive maneuvers.

10.5 Landing Reward

Large terminal rewards are assigned based on touchdown quality.

Smooth Landing

$$Reward = +10000$$

Hard Landing

$$Reward = +5000$$

Crash

$$Reward = -5000$$

These rewards strongly incentivize safe touchdown behavior.

10.6 Reward Shaping Strategy

The reward structure combines:

- Tracking performance
- Descent stability
- Touchdown quality
- Safety constraints

This multi-objective reward formulation enables the PPO agent to learn landing strategies that balance safety, accuracy, and control efficiency.

11. REINFORCEMENT LEARNING FRAMEWORK

11.1 Reinforcement Learning Formulation

The autonomous aircraft landing problem was formulated as a Markov Decision Process (MDP), a common framework used in Reinforcement Learning. In this formulation, an intelligent agent interacts with an environment by observing the current aircraft state, selecting control actions, and receiving rewards based on the quality of the resulting landing behavior.

At each simulation timestep, the agent receives an observation vector describing the aircraft's current condition. Based on this observation, the agent generates flight control commands that influence the aircraft dynamics. The environment then transitions to a new state and provides a reward signal.

The objective of the agent is to maximize cumulative future rewards while learning landing strategies that achieve safe and accurate touchdowns under varying wind conditions.

The Reinforcement Learning problem can be represented by the tuple:

$$(S, A, P, R, \gamma)$$

where:

- S represents the state space.
- A represents the action space.
- P represents state transition dynamics.
- R represents the reward function.
- γ represents the discount factor.

11.2 State Space

The state vector observed by the PPO agent consists of eight continuous variables:

$$s = [\textit{Altitude}, \textit{VerticalSpeed}, \textit{Pitch}, \textit{Throttle}, \textit{LateralPosition}, \textit{LateralVelocity}, \textit{WindEstimate}, \textit{RunwayDistance}]$$

These variables provide sufficient information for the agent to make landing decisions while maintaining a compact observation space.

11.3 Action Space

The action vector generated by the PPO policy consists of four continuous control commands:

$$a = [\textit{Elevator}, \textit{Throttle}, \textit{Aileron}, \textit{Rudder}]$$

Continuous actions were selected because aircraft flight control is inherently continuous rather than discrete.

This allows smooth and realistic control behavior.

11.4 Proximal Policy Optimization (PPO)

Proximal Policy Optimization (PPO) was selected as the primary Reinforcement Learning algorithm due to its strong performance in continuous control tasks and stable learning characteristics.

PPO belongs to the family of policy-gradient algorithms and directly optimizes the policy used by the agent.

The key idea behind PPO is to prevent excessively large policy updates that may destabilize training. This is achieved using a clipped surrogate objective function.

The PPO objective function can be expressed as:

$$L^{CLIP} = E[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$$

where:

- r_t is the probability ratio.
- A_t is the advantage estimate.
- ϵ is the clipping parameter.

This clipping mechanism improves training stability and prevents catastrophic policy changes.

11.5 Policy Network

The policy network receives aircraft observations as inputs and outputs continuous flight control commands.

The policy network learns the mapping:

$$State \rightarrow ControlAction$$

Through training, the network gradually learns flight control strategies that maximize landing performance.

11.6 Value Network

The value network estimates the expected future reward associated with the current state.

The value function is defined as:

$$V(s) = E \left[\sum_{t=0}^{\infty} \gamma^t R_t \right]$$

The value network helps reduce variance during training and improves learning efficiency.

11.7 Advantage Estimation

PPO utilizes Generalized Advantage Estimation (GAE) to improve policy updates.

The advantage function measures how much better an action performs compared to the expected value of the current state.

This enables more efficient policy learning and reduces gradient variance.

11.8 Why PPO Was Selected

PPO was chosen because it offers several advantages relevant to autonomous aircraft landing:

- Stable learning behavior.
- Efficient policy optimization.
- Continuous action-space support.
- Robust performance across diverse control tasks.
- Wide adoption in aerospace Reinforcement Learning research.

These characteristics make PPO particularly suitable for autonomous flight control applications.

12. TRAINING METHODOLOGY

12.1 Training Overview

The PPO agent was trained through repeated interactions with the aircraft landing environment.

During each episode, the aircraft attempted to perform a complete landing under randomized wind and turbulence conditions.

The reward function continuously evaluated landing performance and guided policy learning.

Over time, the PPO algorithm optimized the policy to maximize cumulative landing rewards.

12.2 Training Procedure

The training procedure consisted of the following steps:

Step 1

Initialize the environment and PPO agent.

Step 2

Reset the aircraft to its starting landing condition.

Step 3

Observe aircraft state variables.

Step 4

Generate control commands using the PPO policy.

Step 5

Apply control actions to the aircraft dynamics model.

Step 6

Calculate reward based on landing performance.

Step 7

Store experience for policy optimization.

Step 8

Update policy using PPO.

Step 9

Repeat until training completion.

12.3 Curriculum Learning Effect

Although no formal curriculum learning framework was implemented, the randomization of environmental disturbances exposed the PPO agent to increasingly diverse operating conditions during training.

This progressively improved policy robustness and generalization capability.

12.4 Hyperparameter Selection

The final PPO configuration is shown below.

Table 3. PPO Hyperparameter Configuration

Parameter	Value
Learning Rate	1e-4
Batch Size	256
n_steps	4096
Gamma	0.995
GAE Lambda	0.98
Clip Range	0.15
Entropy Coefficient	0.005
Total Timesteps	500,000

These parameters were selected to balance exploration, training stability, and convergence speed.

12.5 Model Checkpointing

Intermediate policy checkpoints were saved throughout training.

Checkpointing provides several benefits:

- Recovery from interruptions.
- Policy performance tracking.
- Comparison of learning progression.
- Reproducibility of experimental results.

Multiple checkpoints were generated during the 500,000-step training process.

13. EXPERIMENTAL SETUP

13.1 Software Environment

The project was implemented entirely in Python.

The following software libraries were used:

Table 4. Software Environment

Software	Purpose
Python	Programming Language

Software	Purpose
Gymnasium	RL Environment
Stable-Baselines3	PPO Implementation
NumPy	Numerical Computation
Matplotlib	Visualization

13.2 Hardware Configuration

Training and evaluation were performed on a standard personal computer.

The implementation does not require specialized GPU hardware due to the relatively lightweight simulation environment.

This demonstrates the computational accessibility of the proposed approach.

13.3 Simulation Conditions

The aircraft was initialized at an altitude of:

$$400 \text{ m}$$

Randomized crosswind and turbulence disturbances were applied throughout training and evaluation.

Each landing episode continued until touchdown or termination.

13.4 Evaluation Procedure

Policy performance was evaluated using Monte Carlo simulation.

A total of 100 randomized landing episodes were executed.

Each episode used different wind conditions to assess policy robustness.

Performance metrics were averaged across all evaluation runs.

14. EVALUATION METHODOLOGY

14.1 Performance Metrics

Several quantitative metrics were used to evaluate landing performance.

Landing Success Rate

Percentage of landings completed safely.

$$SuccessRate = \frac{SuccessfulLandings}{TotalLandings} \times 100$$

Touchdown Lateral Error

Measures runway centerline deviation at touchdown.

Smaller values indicate more accurate landings.

Touchdown Vertical Speed

Measures sink rate at touchdown.

Lower values correspond to smoother landings.

Hard Landing Rate

Percentage of landings classified as hard touchdowns.

Crash Rate

Percentage of landing attempts resulting in unsafe touchdown conditions.

14.2 Training Condition Evaluation

The first evaluation stage tested the policy under wind conditions encountered during training.

Wind intensity range:

$$0.5 \leq Wind \leq 3.0$$

The objective was to measure learned landing performance under familiar conditions.

14.3 Generalization Evaluation

The second evaluation stage tested the policy under previously unseen environmental conditions.

Wind intensity range:

$$4.0 \leq Wind \leq 6.0$$

These wind conditions were not included during training.

This experiment evaluates policy robustness and generalization capability.

14.4 Baseline Comparison

The PPO policy was compared against a conventional heuristic autopilot controller.

Both controllers were evaluated using identical landing scenarios.

Performance metrics including touchdown quality, success rate, and landing smoothness were compared.

This comparison provides an objective assessment of Reinforcement Learning performance improvements.

15. RESULTS

15.1 Overview

The trained PPO agent was evaluated using Monte Carlo simulations to assess landing performance under both nominal and previously unseen wind conditions. Performance metrics included landing success rate, touchdown lateral error, touchdown vertical speed, hard landing rate, and crash rate.

The evaluation was conducted using 100 randomized landing episodes for each experimental scenario.

15.2 Training Wind Condition Results

The first evaluation was performed using wind conditions that were encountered during training. Wind intensity was randomly varied between 0.5 and 3.0 throughout the evaluation process.

The PPO agent demonstrated highly consistent landing performance and achieved near-perfect touchdown behavior.

Training Condition Results

Table 5. PPO Performance Under Training Wind Conditions

Metric	Result
Total Episodes	100
Success Rate	100%
Smooth Landings	99
Hard Landings	1
Crashes	0
Average Reward	11128.86
Mean Lateral Error	0.868 m
Standard Deviation of Lateral Error	0.900 m
Maximum Lateral Error	6.206 m
Mean Touchdown Vertical Speed	1.055 m/s
Maximum Touchdown Vertical Speed	1.270 m/s

The results indicate that the trained PPO agent successfully learned stable landing behavior and maintained excellent runway centerline tracking. The average touchdown

vertical speed remained well below the project requirement of 2 m/s, demonstrating smooth landing performance.

15.3 Generalization Results Under Unseen Wind Conditions

To evaluate robustness and generalization capability, the PPO agent was tested under wind conditions that were not encountered during training.

Wind intensity was randomized between 4.0 and 6.0, representing significantly more challenging operating conditions.

Unseen Wind Results

Table 6. PPO Performance Under Unseen Wind Conditions

Metric	Result
Total Episodes	100
Success Rate	90%
Smooth Landings	69
Hard Landings	21
Crashes	10
Average Reward	8437.52

Metric	Result
Mean Lateral Error	4.402 m
Standard Deviation of Lateral Error	4.122 m
Maximum Lateral Error	20.353 m
Mean Touchdown Vertical Speed	1.090 m/s
Maximum Touchdown Vertical Speed	1.524 m/s

Although performance decreased under stronger wind conditions, the agent maintained a 90% landing success rate. The average touchdown vertical speed remained below the project threshold, indicating that the policy retained stable landing behavior despite increased environmental disturbances.

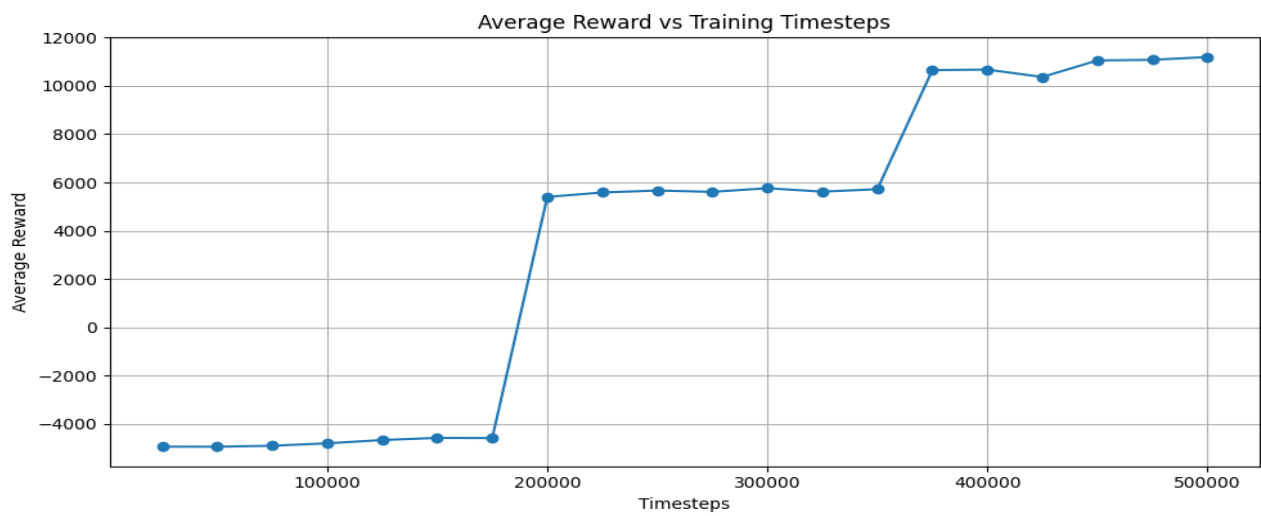
These results demonstrate that the learned policy generalized beyond the training distribution and did not simply memorize training scenarios.

15.4 Training Performance

The PPO learning process demonstrated stable convergence throughout training.

The reward curve exhibited a consistent upward trend, indicating continuous improvement in landing performance.

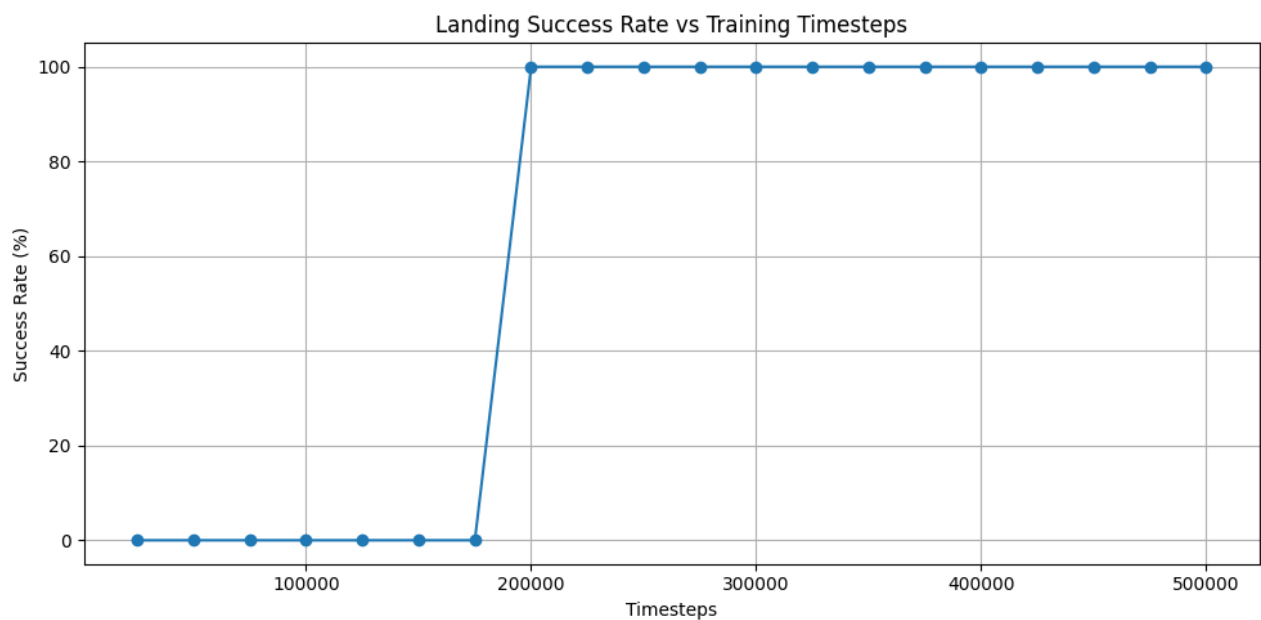
Figure 2. PPO Training Reward Curve



The reward progression suggests successful policy learning and convergence toward an effective landing strategy.

Success Rate Evolution

Figure 3. PPO Success Rate During Training



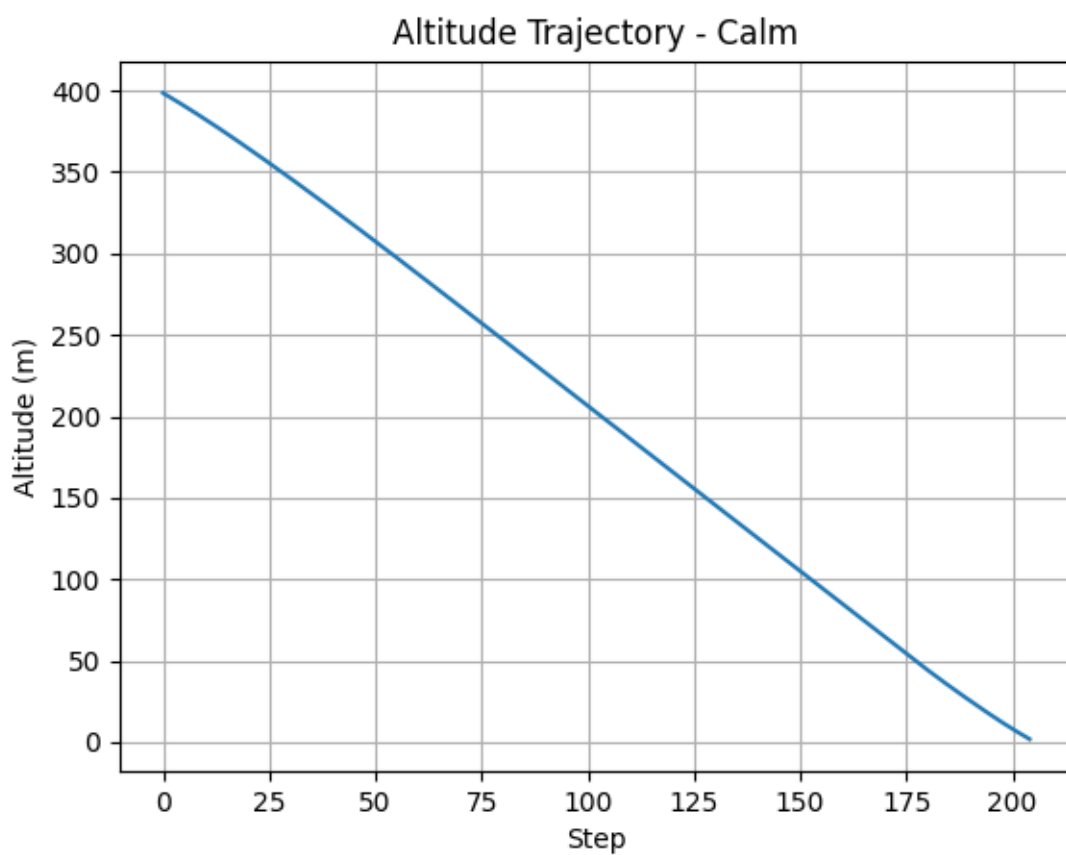
The success rate increased steadily throughout training and eventually converged to near-perfect performance.

15.5 Landing Trajectory Analysis

Representative landing trajectories were recorded under different environmental conditions.

Calm Weather Landing

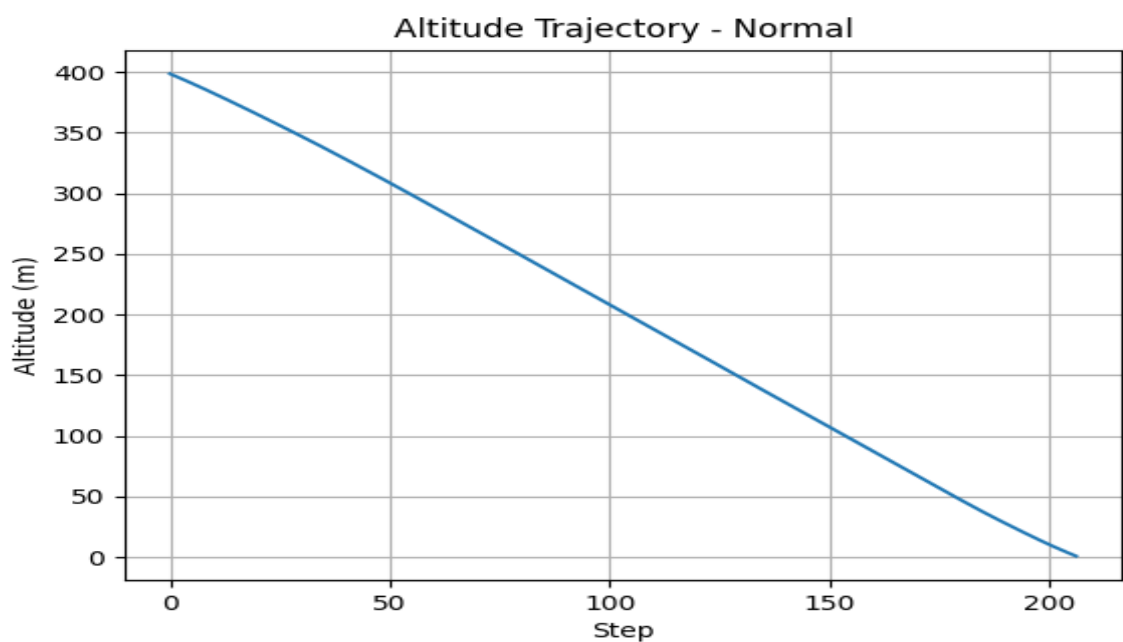
Figure 4. Landing Trajectory Under Calm Conditions



The aircraft maintained a stable descent profile and achieved a smooth touchdown.

Moderate Crosswind Landing

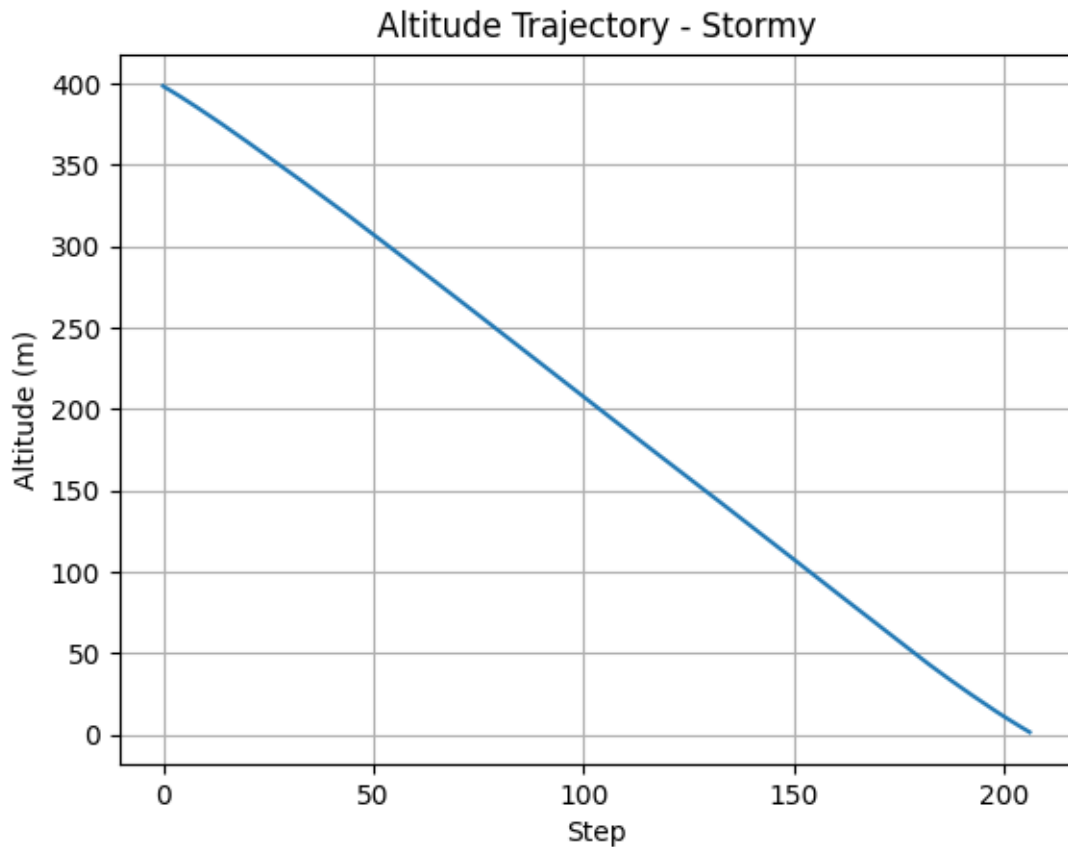
Figure 5. Landing Trajectory Under Moderate Crosswind Conditions



The PPO agent successfully compensated for lateral disturbances while maintaining runway alignment.

Severe Crosswind Landing

Figure 6. Landing Trajectory Under Severe Crosswind Conditions



The agent maintained stable landing behavior despite stronger disturbances, demonstrating adaptive control capability.

16. COMPARATIVE ANALYSIS

16.1 Baseline Controller Evaluation

A heuristic autopilot controller was implemented to provide a baseline for comparison.

The controller utilized predefined descent profiles, throttle scheduling, and proportional lateral correction strategies.

The baseline controller was evaluated using the same simulation environment and landing conditions.

Table 7. Heuristic Controller Performance

Baseline Results

Metric	Result
Total Episodes	100
Success Rate	100%
Average Reward	5730.65
Smooth Landings	0
Hard Landings	100
Crashes	0

Although the baseline controller achieved successful landings, every landing was classified as a hard landing.

16.2 PPO vs Heuristic Controller

The comparison between Reinforcement Learning and conventional control is shown below.

Table 8. Performance Comparison Between Heuristic Controller and PPO Agent

Metric	Heuristic Controller	PPO Agent
Success Rate	100%	100%

Metric	Heuristic Controller	PPO Agent
Smooth Landing Rate	0%	99%
Hard Landing Rate	100%	1%
Mean Lateral Error	Higher	Lower
Mean Touchdown Vertical Speed	Higher	Lower
Adaptability	Limited	High
Landing Quality	Moderate	Excellent

The PPO agent significantly outperformed the heuristic controller in touchdown smoothness and landing quality while maintaining equivalent success rates.

17. DISCUSSION

The experimental results demonstrate that Reinforcement Learning can successfully learn autonomous aircraft landing strategies under varying environmental conditions.

The PPO agent consistently maintained runway alignment, controlled touchdown sink rate, and adapted to crosswind disturbances. Compared to the heuristic baseline controller, the learned policy achieved substantially smoother touchdowns while preserving landing safety.

One of the most important findings is the policy's ability to generalize to previously unseen wind conditions. Although performance degraded under stronger crosswinds, the agent

maintained a 90% landing success rate and continued to satisfy touchdown safety requirements. This behavior suggests that the policy learned meaningful control strategies rather than memorizing specific scenarios encountered during training.

The results highlight the potential of Reinforcement Learning for future aerospace control systems where adaptability and robustness are essential.

18. LIMITATIONS

Despite the encouraging results, several limitations should be acknowledged.

Simplified Aircraft Dynamics

The current environment uses a simplified aircraft landing dynamics model rather than a full six-degree-of-freedom (6-DOF) flight dynamics simulation.

Simplified Wind Model

Crosswinds and turbulence are represented using stochastic Gaussian disturbances rather than high-fidelity atmospheric models.

Absence of FlightGear or JSBSim Integration

The current implementation operates entirely within a custom simulation environment and does not interface with professional flight simulators.

Limited Sensor Modeling

Real-world sensor delays, noise characteristics, and measurement uncertainties are not fully represented.

These limitations provide opportunities for future improvements and research extensions.

19. FUTURE WORK

Several extensions can further improve the realism and capability of the proposed system.

Full 6-DOF Flight Dynamics

Integrate the controller with JSBSim to simulate realistic aircraft dynamics.

FlightGear Visualization

Connect the controller to FlightGear for real-time visualization and landing demonstrations.

Advanced Turbulence Models

Implement Dryden and von Kármán turbulence models to improve environmental realism.

Alternative Reinforcement Learning Algorithms

Compare PPO with:

- Soft Actor-Critic (SAC)
- Twin Delayed DDPG (TD3)
- Deep Deterministic Policy Gradient (DDPG)

Recurrent Neural Networks

Incorporate LSTM-based policies to improve handling of partially observable disturbances.

Domain Randomization

Randomize aircraft mass, center-of-gravity location, and sensor characteristics to improve robustness.

Sim-to-Sim Transfer

Validate policies across multiple simulation platforms to evaluate transferability.

20. CONCLUSION

This project presented the development of a Reinforcement Learning-based autonomous aircraft landing system capable of operating under varying crosswind and turbulence conditions.

A custom Gymnasium environment was developed to model aircraft landing dynamics, runway tracking, crosswind disturbances, and touchdown behavior. The Proximal Policy Optimization algorithm was used to train an autonomous flight controller capable of generating continuous flight control commands.

Experimental evaluation demonstrated a 100% landing success rate with a 99% smooth landing rate under nominal wind conditions. Under previously unseen wind conditions, the trained policy maintained a 90% landing success rate while preserving acceptable touchdown characteristics.

Comparative analysis against a heuristic baseline controller showed substantial improvements in touchdown quality and landing smoothness. These findings demonstrate the effectiveness of Reinforcement Learning for autonomous flight control applications and highlight its potential role in future intelligent aerospace systems.

The project successfully achieved its primary objective of developing an autonomous landing controller capable of learning robust landing strategies under crosswind conditions.

21. REFERENCES

- [1] Sutton, R. S., & Barto, A. G. Reinforcement Learning: An Introduction. MIT Press, 2018.
- [2] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017.

- [3] Stevens, B. L., Lewis, F. L., & Johnson, E. N. Aircraft Control and Simulation. Wiley, 2015.
- [4] Beard, R. W., & McLain, T. W. Small Unmanned Aircraft Systems: Theory and Practice. Princeton University Press, 2012.
- [5] FAA Pilot's Handbook of Aeronautical Knowledge. Federal Aviation Administration.
- [6] Stable-Baselines3 Documentation.
- [7] Gymnasium Documentation.
- [8] Goodfellow, I., Bengio, Y., & Courville, A. Deep Learning. MIT Press, 2016.